

Marque — Product Requirements Document

Version: 1.0 **Status:** Ready for Development **Author:** Product **Last Updated:** April 2026
Audience: Engineering, Design, Investors

Document Control

Field	Detail
Product Name	Marque
Platform	iOS (iPhone first)
Target Launch	v1.0 MVP
Prototype Status	Local garage app fully built; cloud/social layer to be developed
Stakeholders	Founder, Design, Engineering, QA

Section 1 — Product Summary

Marque is an iOS app that helps car owners store, track, and share their vehicle information. It solves the universal problem of scattered car records — insurance cards in glove boxes, service receipts in drawers, expiry dates forgotten in emails — by giving every car a permanent, organized home on your phone.

For car enthusiasts, Marque goes further: it lets them build a public garage, follow other collectors, and showcase the vehicles they're proud of.

The app works for two audiences simultaneously. A Toyota Camry owner who just wants to remember when their registration expires. A BMW collector with six cars who wants an audience for their fleet. Both use the same product and neither compromises the other's experience.

A functional prototype exists covering the core garage experience: adding cars by VIN or

manually, tracking maintenance and service history, storing photos, calculating expenses, and scheduling expiry reminders. The v1.0 PRD covers the full product including authentication, cloud sync, and the social layer.

Section 2 — User Personas

Persona 1: The Practical Owner

"I just want to know when my insurance expires and what my mechanic did last time."

- **Name:** Marcus, 34
 - **Occupation:** Software engineer
 - **Cars:** 1 daily driver (Honda Accord)
 - **Pain point:** Keeps insurance and registration info in his email. Loses service receipts. Missed a registration renewal last year.
 - **What he wants:** A simple, private record of everything about his car. No friction, no social pressure.
 - **What he doesn't want:** Notifications he didn't ask for. Apps that upsell him every screen.
 - **Platform behavior:** Checks the app a few times a year, mostly around renewal dates.
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Persona 2: The Enthusiast

"I've owned 11 cars. I want to show my collection and find people who understand why I have three of them."

- **Name:** Sofia, 27
- **Occupation:** Freelance photographer
- **Cars:** 3 currently (Porsche 944, Miata, daily BMW)
- **Pain point:** Posts car content on Instagram but it's lost in her general feed. No dedicated place for her car identity. Can't find local enthusiasts by what they drive.
- **What she wants:** A garage that feels like a portfolio. A feed of people who share her taste.
- **What she doesn't want:** A generic social feed full of noise. A product that feels like a DMV form.

- **Platform behavior:** Opens the app several times a week. Adds photos, browses Explore, follows new accounts.

Persona 3: The Upgrader

"I track everything about my cars. I need all of this synced across my iPhone and iPad, and I want to export service history when I sell."

- **Name:** David, 42
- **Occupation:** Contractor and car restorer
- **Cars:** 5 (3 project cars + 2 dailies)
- **Pain point:** Keeps spreadsheets. Wants everything in one place across devices. Buyers want to see full service history when purchasing.
- **What he wants:** Unlimited cars, PDF export, cloud sync, reliability.
- **What he doesn't want:** Apps that lose his data or reset between updates.
- **Platform behavior:** Power user. Will hit the free tier limit. Natural Pro subscriber.

Section 3 — Problem Statement

The core problem

There is no purpose-built app for managing personal vehicle information that is simultaneously useful enough for a practical owner and compelling enough for an enthusiast.

Current workarounds and why they fail:

Workaround	Why It Fails
Notes / spreadsheets	No structure, no reminders, no sharing
Insurance company apps	One car, one company, no maintenance tracking
General garage apps (CarFax, etc.)	Dealer-centric, not owner-centric; ugly, no social layer
Instagram / Facebook	General social feed; no car-specific organization; no private records
Paper / glove box	Lost, damaged, inaccessible remotely

The specific gaps Marque fills

1. **Centralized, structured storage** for every piece of car information a person cares about — specs, service records, photos, insurance, registration.
2. **Proactive reminders** before things expire, without the user having to remember to check.
3. **A public identity layer** that lets enthusiasts show off their collection without polluting a general social feed.
4. **A community discovery tool** so enthusiasts can find and follow people based on what they actually drive.

Section 4 — Goals and Non-Goals

Goals — v1.0

#	Goal	Metric
G1	Give any car owner a better place to store vehicle info than their email/notes	1,000 cars added in first 30 days
G2	Make expiry reminders reliable and actionable	>50% of users with expiry dates enable notifications
G3	Create a social layer that enthusiasts want to share	20% of users make at least one car public within 7 days of signup
G4	Establish a monetization path that doesn't compromise free-tier value	>3% conversion to Pro within 90 days of launch
G5	Deliver a crash-free, fast experience	>99.5% crash-free sessions, <2s cold launch

Non-Goals — v1.0

These are explicitly deferred. Any feature request that falls under these should be rejected for v1.

Non-Goal	Why Deferred
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Direct messaging / chat	High complexity, moderation liability, distraction from core
Fuel / mileage logbook	Valuable but a separate use case; validate core first
Document storage (PDFs, receipts)	Storage costs, camera UX complexity; v2
Car valuation / market pricing	Third-party API cost, legal liability; v2
Android version	Platform focus required for quality; v2
Web app or dashboard	iOS-first strategy; v2
Marketplace / classifieds	Completely different product; v3
Dealer / business accounts	B2B is a different sales motion; future
Comments on public cars	Moderation burden, distraction from core; v1.5
Algorithmic feed	No data to train on at launch; chronological for v1

Section 5 — Key Features

Organized by priority tier:

Tier 1 — Core Garage (Prototype Complete)

Feature	Description	Status
VIN Decode	Add a car by entering its 17-digit VIN; fields auto-populate via NHTSA API	Built
Manual Entry	Add a car by picking make from a list and entering model/year	Built
Car Detail	Full record per car: specs, photo, registration, insurance, notes	Built
Maintenance Log	Log service records (type, date, mileage, cost, shop, notes)	Built
Expense Tracking	Aggregated cost view by vehicle and category, filterable by time period	Built
Expiry Alerts	Push notifications 30 days, 7 days, and on the day for insurance and registration expiry	Built

Car Photo	Add, change, reposition (drag), and remove car photos	Built
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Tier 2 — Cloud + Identity (v1.0 New)

Feature	Description	Priority
Authentication	Sign in with Apple, Google, and Email; 2FA via Firebase	P0
Cloud Sync	All cars, records, and photos synced to Firestore and Firebase Storage	P0
Offline Mode	Full read/write access with auto-sync when connection restores	P0
User Profile	Username, display name, profile photo, bio, public stats	P0
Profile Setup	Post-signup onboarding: username, photo, bio	P0
Local Data Import	Migrate existing prototype data to cloud on first login	P1
Pro Tier	Unlimited cars, PDF export, Pro badge via monthly/annual subscription	P1

Tier 3 — Social Layer (v1.0 New)

Feature	Description	Priority
Public/Private Toggle	Per-car visibility control, default private	P0
Public Profile	Read-only view of another user's public garage	P1
Follow / Unfollow	Follow other users; see their new cars in a following feed	P1
Explore	Browse public garages, search by make/model/username	P1
Following Feed	Chronological list of new cars from followed users	P1

Section 6 — User Stories

Authentication

ID	Story	Acceptance Criteria
US-01	As a new user, I want to sign up with Apple ID so that I don't need to create a new password	Auth completes in <3 taps with no form entry
US-02	As a new user, I want to sign up with Google so that I can use my existing account	Google sheet appears, auth completes on return
US-03	As a new user, I want to sign up with email so that I have an account independent of Apple/Google	Email verified before full access granted
US-04	As a returning user, I want to be automatically signed in so that I don't have to log in every time	Auto sign-in on launch if token valid
US-05	As a user who forgot my password, I want to reset it via email	Reset email delivered; new password works immediately

Garage

ID	Story	Acceptance Criteria
US-06	As a car owner, I want to add my car by VIN so that I don't have to type all the specs manually	VIN search auto-populates make, model, year, trim, engine, fuel, transmission within 3s
US-07	As a car owner, I want to add my car manually when I don't have the VIN	Manual path requires only make, model, year to save
US-08	As a car owner, I want to add a photo to my car and reposition it	Photo picker opens with permission request; drag gesture repositions
US-09	As a car owner, I want to log a service record with cost, shop, and notes	Form accepts all fields; service type is required; all others optional
US-10	As a car owner, I want to see my total maintenance spend over time	Expense view shows totals by vehicle and by category, filterable by period
US-11	As a car owner, I want to be reminded before my insurance and registration expire	Notifications fire at 30 days, 7 days, and on the day; tap goes to that car
US-12	As a car owner, I want my data accessible even when I'm offline	App shows full garage from local cache when offline

US-13	As a user with existing local data, I want to import it to my account	Import prompt on first login; all cars and records transferred
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Social

ID	Story	Acceptance Criteria
US-14	As an enthusiast, I want to make individual cars public so that others can see them	Per-car toggle; changes visibility immediately; default is private
US-15	As a user, I want a profile page others can visit to see my public cars	Profile shows avatar, bio, stats, and public car grid
US-16	As a user, I want to follow others so that I can see their new cars	Follow/unfollow from public profile; following feed updates accordingly
US-17	As a user, I want to discover new garages in Explore	Explore shows public cars, searchable by make/model/username
US-18	As a user, I want to view another user's public car without seeing their private info	VIN, license plate, insurance details, costs never shown on public view

Pro

ID	Story	Acceptance Criteria
US-19	As a user with 4+ cars, I want to upgrade to Pro to add more	Paywall shown on 4th car add; purchase unlocks immediately
US-20	As a Pro user, I want my subscription restored on a new device	Restore purchase link on paywall; Firestore updated on restore

Section 7 — Functional Requirements

FR-01: Authentication

- **FR-01.1** App must support Sign in with Apple, Google, and Email/Password
- **FR-01.2** Sign in with Apple must be listed first (App Store requirement when offering any social login)

- **FR-01.3** Email accounts must verify their address before full access is granted
- **FR-01.4** Firebase Auth token must be refreshed silently; the user should not be prompted to re-authenticate within a 90-day session
- **FR-01.5** If the same email exists under two providers, the app must offer to link the accounts rather than show an error
- **FR-01.6** Profile setup (username, photo, bio) must be completed before the main app is accessible; only username is required
- **FR-01.7** Usernames must be 3–30 characters, alphanumeric and underscores only, globally unique
- **FR-01.8** Username availability must be checked in real-time with a 500ms debounce

FR-02: Car Management

- **FR-02.1** VIN decode must call NHTSA VPIC API; response must populate make, model, year, trim, body style, drive type, engine, fuel type, and transmission
- **FR-02.2** VIN input must be validated as exactly 17 characters before the search button enables
- **FR-02.3** If VIN decode fails or returns no data, the user must be able to proceed with manual entry without re-opening the flow
- **FR-02.4** Minimum required fields to save a car: make, model, year
- **FR-02.5** All car records must be stored in Firestore under `users/{userId}/cars/{carId}`
- **FR-02.6** Car deletion must delete: the Firestore document, all maintenance subcollections, and the photo from Firebase Storage
- **FR-02.7** Car data must be readable offline via Firestore's local persistence cache

FR-03: Photos

- **FR-03.1** The app must request `PHPhotoLibrary` authorization before opening the picker; never at app launch
- **FR-03.2** If authorization is denied, the app must show an alert with a direct link to iOS Settings
- **FR-03.3** Photos must be stored in Firebase Storage at `users/{userId}/cars/{carId}/photo.jpg`
- **FR-03.4** Photo upload must not block saving other car fields; it must proceed in the background
- **FR-03.5** The user must be able to drag the photo vertically to set a crop offset, stored as `photoOffsetY`
- **FR-03.6** Photo removal must be confirmed via a confirmation dialog before deletion

- **FR-03.7** Photo deletion from Firebase Storage must occur immediately on user confirmation, not deferred to save

FR-04: Notifications

- **FR-04.1** Notification permission must be requested when the user first sets any expiry date, not on first app launch
- **FR-04.2** The app must schedule local notifications at 30 days, 7 days, and on the day of expiry for both insurance and registration
- **FR-04.3** Each notification must carry a deep link payload containing the car ID
- **FR-04.4** Tapping a notification must navigate directly to the relevant car's detail screen
- **FR-04.5** When a car's expiry date is updated, all existing notifications for that car must be cancelled and rescheduled
- **FR-04.6** When a car is deleted, all its associated notifications must be cancelled

FR-05: Cloud Sync

- **FR-05.1** Firestore offline persistence must be enabled with a minimum 100MB cache
- **FR-05.2** Writes while offline must queue and sync automatically when connectivity is restored
- **FR-05.3** The UI must indicate offline status via a non-blocking banner; it must not prevent the user from reading or creating records
- **FR-05.4** Photo uploads that fail due to network conditions must be retryable without re-selecting the photo

FR-06: Public / Private Visibility

- **FR-06.1** Every car must default to private (`isPublic = false`) at creation
- **FR-06.2** The user must be able to toggle individual cars public or private from the Car Detail screen
- **FR-06.3** A public car must never expose: VIN, license plate, insurance provider, insurance policy number, insurance expiry date, registration expiry date, or individual maintenance costs
- **FR-06.4** A public car must expose: make, model, year, color, mileage, trim, body style, drive type, engine, fuel type, transmission, photo, notes (if present), and service types + dates (without costs)
- **FR-06.5** Visibility changes must take effect in Firestore immediately on toggle

FR-07: Social — Follow

- **FR-07.1** Any authenticated user can follow any other user from their public profile
- **FR-07.2** Follower and following counts must update optimistically on action and confirm on Firestore write
- **FR-07.3** The following feed must show new public cars added by followed users in reverse chronological order
- **FR-07.4** Unfollow must require confirmation: "Unfollow {username}?"
- **FR-07.5** A user must be able to block another user; a blocked user cannot view the blocker's profile or cars

FR-08: Pro Tier

- **FR-08.1** Free tier is capped at 3 cars; the 4th add attempt surfaces the upgrade sheet
- **FR-08.2** The upgrade sheet must offer both monthly (\$2.99) and annual (\$24.99) plans; annual shown first as recommended
- **FR-08.3** Pro must be verified via Firestore (`user.isPro = true`), not StoreKit alone, to support cross-device unlock
- **FR-08.4** When a subscription lapses, existing cars must remain readable; only new car additions are blocked
- **FR-08.5** "Restore purchase" must be visible on the upgrade screen without scrolling
- **FR-08.6** The app must handle App Store server-to-server notifications for subscription events (renewal, cancellation, billing failure) via Firebase Cloud Functions

Section 8 — Edge Cases

ID	Scenario	Expected Behavior
EC-01	User adds a car, immediately goes offline before Firestore write completes	Write queued; car shows in UI immediately; syncs on reconnect; no data lost
EC-02	Two devices logged into same account simultaneously add a car	Firestore last-write-wins; both cars appear; no conflict
EC-03	VIN decode returns partial data (some fields missing)	Available fields pre-filled; empty fields left blank for manual entry

EC-04	User adds 3 cars, hits paywall, purchases Pro, then cancels subscription within 7 days (Apple refund)	Firestore updated via server notification; free limit reinstated; existing 3 cars unaffected
EC-05	User attempts to sign up with email already linked to an Apple ID	"An account with this email exists. Sign in with Apple instead?" prompt with Apple button
EC-06	Photo upload fails after car is already saved	Car saved without photo; car detail shows "Photo didn't upload. [Retry]"
EC-07	User sets a registration expiry date in the past	Notification scheduled for the past silently dropped; expiry banner shows "Expired" in UI
EC-08	User deletes account with 3 public cars	All Firestore documents deleted; Firebase Storage files deleted; follow relationships removed; public profile becomes 404
EC-09	User makes a car public then immediately deletes it	Car removed from Explore feed; anyone who bookmarked a direct link sees "This car is no longer available"
EC-10	VIN search returns a result for a car with a non-English make name	Normalized against the <code>CarData.makes</code> list; falls back to raw API value if no match
EC-11	User taps notification for a car that has been deleted	Deep link resolves; car not found; user lands on My Garage with no error message
EC-12	User reinstalls the app with an active Pro subscription	Firebase Auth restores session; Firestore <code>isPro</code> flag restores Pro features; no re-purchase required
EC-13	User opens app in airplane mode on first install	Auth cannot complete; "You need an internet connection to create your account"
EC-14	User enters a duplicate username during profile setup	Real-time check shows "taken" before they submit; suggestions offered
EC-15	User uploads a very large photo (12MP+ iPhone camera)	Photo compressed to JPEG at 0.8 quality before upload; original not stored

Section 9 — Success Metrics

Acquisition

Metric	Target (30 days post-launch)
App Store installs	2,500
Signup completion rate (install → account created)	>60%
Auth method distribution	Apple >50%, Google >30%, Email <20%

Activation

Metric	Target
Time to first car added	<3 minutes from signup
D1 activation (adds at least 1 car on day 1)	>50% of signups
Photo upload rate	>40% of cars have a photo within 7 days
Notification opt-in rate	>50% of users who set an expiry date

Engagement

Metric	Target (Day 30)
D7 retention	>35%
D30 retention	>20%
Cars per active user	>1.8
Maintenance records per active car	>2
Public car rate	>20% of users make at least one car public
Follow rate (users who follow at least one person)	>15%

Monetization

Metric	Target (90 days)
Paywall conversion rate	>3% of active free users

Annual vs monthly split	>60% annual
MRR	\$500 (proves model; not a scale target)
Churn rate (monthly Pro)	<8%/month

Quality

Metric	Target
Crash-free session rate	>99.5%
App Store rating	>4.3 stars
Cold launch time	<2 seconds
Firestore sync success rate	>99.9%

Section 10 — Launch Risks

ID	Risk	Likelihood	Impact	Mitigation
R-01	Low activation — users sign up but don't add a car	High	High	Empty state with clear CTA; onboarding ends at Add Car, not at Garage
R-02	Social layer feels empty at launch (no one to follow)	High	Medium	Pre-seed with founder accounts and curated public garages
R-03	Firebase costs spike unexpectedly	Medium	High	Set Firebase budget alerts at \$50/month; enforce photo compression; cache aggressively
R-04	App Store rejection for Sign in with Apple parity	Medium	High	Apple requires Apple login to be equal prominence; design per HIG
R-05	NHTSA API downtime breaks Add Car flow	Low	High	Fallback message with manual entry path always available

R-06	StoreKit purchase loop or double-charge	Low	High	Test all StoreKit edge cases with Sandbox; idempotent purchase handling
R-07	User data loss on UserDefaults → Firestore migration	Low	Very High	Import is additive only; never delete UserDefaults until Firestore write confirmed
R-08	Content moderation exposure (public profiles)	Medium	Medium	Block and report flows from day one; Firebase App Check; ToS prohibiting illegal content
R-09	Negative reviews from confusing UX for non-enthusiasts	Medium	Medium	Normal users should never see social features until they choose to
R-10	GDPR / CCPA compliance gap	Low	High	Privacy policy on file before launch; "Delete Account" must actually delete all data

Section 11 — MVP Definition

What is in v1.0 MVP

Must ship:

- Sign in with Apple, Google, Email (+ email verification)
- Profile setup (username required; photo, bio optional)
- Local data import (for existing prototype users)
- Full garage experience: add by VIN/manual, edit, delete, photo, maintenance log, expense view
- Expiry notifications with deep links
- Firestore cloud sync + offline mode
- Public/Private per-car toggle
- Public profile page (read-only)
- Follow / Unfollow
- Following feed (chronological)
- Explore tab (public garages, basic search)
- Paywall at 3-car limit + Pro subscription (monthly + annual)

- Block and report (required for App Store social features approval)
- Privacy policy + Terms of Service in-app

Cut from v1.0, ship in v1.1:

- Comments on public cars
- PDF export (teased as Pro feature, delivered in v1.1)
- Likes / reactions
- Following feed notifications
- Follower/following list screens

Definition of Done for MVP

The MVP is complete when:

1. A new user can sign up, add a car by VIN, receive an expiry notification, and make their car public in under 5 minutes on a production build
2. A returning user can open the app offline and see their full garage
3. A user with 4 cars is prompted to upgrade and can complete a purchase
4. The app passes App Store review and is live in the App Store
5. Crash-free rate is above 99% on first 48 hours of production traffic

Appendix A — Out-of-Scope Features (Parking Lot)

Feature	Target Version	Notes
DM / Chat	v2.0	Requires moderation; complex push infra
Fuel logbook	v2.0	High-frequency use case; justified after retention proven
Document storage	v2.0	Storage costs; camera UX complexity
Car valuation	v2.0	Third-party data licensing cost
Android	v2.0	After iOS PMF confirmed

Marketplace	v3.0	Different product; different legal surface
Business / dealer accounts	v3.0	B2B motion; different GTM
Apple Watch	v3.0	Nice-to-have; low user value at this stage

Appendix B — Assumptions

1. Firebase free tier is sufficient for the first 1,000 active users without billing
2. NHTSA VPIC API remains free and publicly available without an API key
3. Apple's App Store review will not flag the social features if block/report is implemented
4. Users will accept a 3-car free limit without significant churn if the value per car is high enough
5. The target demographic (car owners in the US) is sufficiently iOS-dominant to justify iOS-first

Appendix C — Glossary

Term	Definition
VIN	Vehicle Identification Number — 17-character unique vehicle code
NHTSA	National Highway Traffic Safety Administration — US agency providing the free VIN decode API
Firestore	Google Firebase's NoSQL cloud database
Firebase Storage	Google Firebase's file storage service
isPublic	Per-car boolean field controlling whether a car appears on the user's public profile and in Explore
isPro	Per-user boolean field in Firestore controlling Pro feature access
photoOffsetY	Stored vertical offset for the car photo crop position

Deep link	A URL or notification payload that opens the app to a specific screen
Expiry banner	The red/orange warning shown in Car Detail when insurance or registration is expired or expiring
Activation	A user has added at least one car to their account

This document represents the complete product requirements for Marque v1.0. It supersedes all prior informal discussions, wireframes, and planning notes. Any scope changes require a documented revision with version bump.

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